FragGen V4 — Final Project Summary

File-fragment type classification for digital forensics CSE4074 Forensics Analytics · repository cleanup and research completion · 2026-09-15

This is the closing record of the repository-wide refactor: what the audit found, what moved where, what the completed ablation study shows, and what is needed to treat this as a finished, presentable, reproducible project.

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1. What changed

phase	outcome
1. Audit	106,708 files / 51.3 GiB inventoried before any change. `docs/REPOSITORY_AUDIT.md`
2. Reorganisation	Moved to the target layout below. Every code/report/checkpoint move verified by SHA-1; every bulk-data folder verified by file count and total bytes
3. Cleanup	65 files deleted (caches, byte-duplicate JSON, one 0-byte failed-run log, superseded demo outputs); 2,194 files archived, byte-identical, to `archive/`. Nothing deleted outside those categories
4. Ablation	The B_aug → E_full ladder completed on Dataset V4 (12 runs, 3 seeds × 4 configs), joining the existing A_plain runs
5. Figures	8 publication figures, matplotlib only
6. Explainability	One inference module (`src/inference`); JSON/CSV reports and PDF stream mapping in `src/explainability`
7. Demo	CLI, Streamlit app and viva script, all calling that one module; a real header-exclusion bug found and fixed during consolidation
8. Documentation	Demo guide, viva script, 15-question examiner Q&A
9. Report	`docs/thesis/RESULTS_AND_DISCUSSION.md` (+ PDF)
10. This check	Tests, links, demo, figures and this summary

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2. Final folder tree

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FragGen/
├── README.md · LICENSE · requirements.txt · .gitignore
├── src/            models · training · inference · preprocessing · explainability · utils
├── configs/        default.yaml, real.yaml
├── data/
│   ├── raw/            corpora (git-ignored; data/raw/README.md)
│   ├── processed_v4/   train.npz · val.npz · test.npz
│   └── metadata/       v4_split_manifest.csv · v4_fragment_metadata.csv · v4_build_report.md
├── checkpoints/
│   ├── v4/             15 checkpoints (A_plain, B_aug, C_aux, D_adv, E_full × 3 seeds)
│   └── calibration/     A_plain_s1337.calib.json
├── reports/
│   ├── figures/        8 figures (+ cross_corpus/, 5 more from the August study)
│   ├── tables/         8 CSV/Markdown result tables
│   ├── confusion_matrices/  21 pooled + long-format confusion-matrix files
│   ├── ablations/       v4/ (15 runs) and cross_corpus_fft75_govdocs1/ (validated August results)
│   └── datasets/        DATA_AUDIT_2026-09-14.md, DATASET_V4_AND_CLEANUP_2026-09-14.md
├── docs/
│   ├── thesis/          RESULTS_AND_DISCUSSION (.md + .pdf), FINDINGS.md, REVIEW1.md
│   ├── viva/            DEMO_GUIDE.md, VIVA_SCRIPT.md, EXAMINER_QA.md
│   ├── architecture/    ARCHITECTURE.md
│   ├── REPOSITORY_AUDIT.md (+ inventory.csv), REMOVED_FILES.md, DATASETS.md
│   └── FINAL_PROJECT_SUMMARY.md (this file, + .pdf)
├── demo/           cli/ (demo.py, viva_demo.py) · streamlit/app.py · samples/ (11 files) · outputs/
├── scripts/        build, train-driver, analysis, calibration, figures, PDF rendering, link check
├── tests/          26 unit + integration tests
└── archive/        2,194 files, byte-identical, superseded datasets/checkpoints/code (archive/README.md)
```

3. Files removed and archived

Full itemised record: `REMOVED_FILES.md`.

outcome	files	size
deleted (caches, duplicates, a failed-run log, an empty marker)	52	282.4 KiB
deleted (superseded demo outputs, once their replacements existed)	13	393.3 KiB
archived, byte-identical	2,194	6.3 GiB
relocated inside the active tree, nothing removed	104,422+	44.8 GiB
empty folders removed	14	—

Nothing under `data/`, `checkpoints/` or any results folder was deleted — only moved to `archive/` or renamed within the active tree. The two datasets superseded by V4 (V2, V3) and the FragmentSets behind the cross-corpus study are archived, not deleted, because they back a validated result and remain reproducible.

4. Research completed

4.1 Ablation study (`reports/ablations/v4/ABLATION_RESULTS.md`)

The cumulative ladder — A_plain, B_aug, C_aux, D_adv, E_full — was trained on Dataset V4 under an identical protocol (6 epochs, seeds 1337/1338/1339, 4096 B, every training fragment), the same driver and settings as the existing A_plain runs.

config	test accuracy	Δ vs A_plain	Holm p	McNemar seeds significant
A_plain	0.8938 ± 0.0036	—	—	—
B_aug	0.8958 ± 0.0032	+0.0019	1.000	2/3
C_aux	0.8938 ± 0.0086	−0.0001	1.000	2/3
D_adv	0.8603 ± 0.0152	**−0.0336**	0.313	**3/3**
E_full	0.8760 ± 0.0159	−0.0179	0.759	2/3

No mechanism improves test accuracy at a level distinguishable from seed-to-seed variation at n = 3. The entropy adversary (D_adv) shows the same behaviour it showed in the earlier 12-seed cross-corpus study on an unrelated dataset: its McNemar test is significant in every seed, and in none of them does it fix more fragments than it breaks — a consistent, if statistically unconfirmed at this sample size, cost. E_full also costs roughly 7× A_plain's training time for a directionally worse result. Full statistics, per-class breakdowns and the run provenance (including a GPU memory-contention issue mid-run and its allocator-only fix) are in that folder.

Consistency checks passed: all 15 checkpoints were re-scored independently on the GPU and matched their original evaluation reports exactly (max |Δ| = 0); accuracy and macro-F1 recomputed from each confusion matrix matched the reported values to floating-point precision.

4.2 Figures (`reports/figures/`, matplotlib only)

figure	shows
fig1_confusion_matrix	A_plain, V4 test, 3 seeds pooled
fig2_precision_recall	per-class precision/recall and F1
fig3_fragment_confidence_timeline	live inference on 3 samples
fig4_dataset_class_distribution	V4 balance, by split and by source
fig5_training_curves	selection-split accuracy per epoch, all 5 configs; E_full pre-training loss
fig6_ablation_comparison	val/test accuracy and macro-F1 per config, with significance annotations
fig7_pdf_container_analysis	one file's stream map + all 1,500 test PDF fragments by stream type
fig8_pipeline_architecture	pipeline + ByteCNN + ablation mechanisms (also SVG)

4.3 Reports and tables

`reports/ablations/v4/ABLATION_RESULTS.md`, `reports/tables/v4_*.{csv,md}` (8 files), `reports/confusion_matrices/v4_*.csv` (21 files, pooled and per-run), and `reports/tables/v4_test_pdf_fragments_by_stream_region.{csv,json}` — the byte-verified PDF stream-type breakdown behind Section 9 of the results report.

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5. Reproducibility checklist

- [x] No broken imports — `tests/test_repository.py` imports every `src` module and compiles all active code
- [x] No import of the pre-refactor `fraggen` package anywhere in `src/`, `scripts/`, `demo/`, `tests/`
- [x] No duplicate files — the 5 byte-identical duplicates found in the audit were removed; the rest are documented as intentional (demo samples) or genuine corpus duplicates
- [x] All reports linked correctly — `python scripts/check_links.py` finds 0 broken links in 28 active documents (1 legacy pre-refactor path noted, harmless)

- [x] All figures render — 8/8 regenerated from current data, visually inspected
- [x] Streamlit launches — confirmed live in-browser: sample selection, verdict banner, vote chart, confidence chart, fragment timeline with PDF stream map, per-fragment table and downloads all work
- [x] CLI works — `demo/cli/demo.py` tested on multiple samples including the tricky PDF, with `--map`
- [x] Viva script runs — full run including the stream-map explanation, `--no-pause` mode verified
- [x] Checkpoints load — all 15 V4 checkpoints load with the correct 9-class label space (`tests/test_checkpoints.py`)
- [x] README is accurate — quick start, layout, reproduction commands and ablation table match the actual repository and results
- [x] Test suite passes — 26/26 (`python -m unittest discover -s tests -t .`)
- [x] Refactor equivalence — the new `src` package verified bit-identical to the pre-refactor `fraggen` package (logits, evaluation output, one-epoch training weights) before the old package was retired

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6. GitHub readiness checklist

- [x] `.gitignore` excludes raw data, `archive/data`, `archive/checkpoints`, `demo/outputs`, and Python/OS junk
- [x] `LICENSE` (MIT) covers code and documentation; `docs/DATASETS.md` and the README state that the datasets are not covered and carry their own third-party terms
- [x] `requirements.txt` is current (adds `streamlit`, notes tested versions)
- [ ] **Repository is not yet a git repository.** `git init`, then an initial commit, is needed before pushing
- [ ] `data/processed_v4/train.npz` is 132 MiB, over GitHub's 100 MB per-file limit — track with Git LFS, or keep it out of the pushed repository and rely on `data/metadata/` + `scripts/build_processed_v4.py` to rebuild it
- [ ] Decide whether `checkpoints/` (204 MiB across 15 files) goes into the repository, a release asset, or Git LFS
- [x] No secrets or machine-specific absolute paths in tracked files (`.claude/` and `powershell.cmd` are local environment files and are git-ignored)

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7. What is intentionally out of scope

Per the cleanup brief's own rules, this pass did not: retrain any deployed model, alter any trained checkpoint's weights, overwrite the validated cross-corpus result, or invent any metric not read from an experiment output file. Every number in `docs/thesis/RESULTS_AND_DISCUSSION.md` traces to a named file in `reports/` or `checkpoints/`.

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8. One remaining decision for the user

`git init` was not run — creating a git repository is an environment change outside this cleanup's remit. Once initialised, the two size issues above (`train.npz` and the checkpoints) should be resolved before the first push.